

DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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Abstract—Adversarial imitation learning reduces manual reward engineering by learning policy rewards from a discriminator. However, when discriminator scores are reused as rewards, the discriminator is no longer merely a classifier: its sensitivity to input variations and the numerical scale of its outputs jointly determine the shape and strength of the policy learning signal. These properties can vary substantially across motions and objectives, causing a fixed discriminator configuration to produce inconsistent reward shaping even when classification remains stable. We present DARE, a differential adversarial reward formulation for reference-conditioned humanoid motion tracking that decouples critic geometry from reward calibration. DARE applies spectral normalization throughout the differential discriminator to bound its sensitivity to tracking residuals, while independently calibrating the reward-producing score scale from residual statistics observed during training. This separation enables a single configuration to be reused across motions without motion-specific reward or discriminator tuning. We evaluate DARE on six humanoid skills with diverse dynamics and contact patterns. DARE outperforms DeepMimic, AMP, and ADD on all four macro-averaged geometric tracking metrics, while achieving reference-consistent foot progression and the lowest lateral drift. These results highlight critic geometry and reward calibration as distinct design considerations for learned imitation rewards.

I. INTRODUCTION

Physics-based humanoid motion imitation requires the controller to reproduce a prescribed reference motion under dynamic constraints, actuator limits, and complex contact conditions. Whether an imitation succeeds is not a single geometric question. The root must translate and turn as the reference does, the articulated body must maintain consistent poses, joint and root velocities must stay synchronized with the reference motion, and contacts must occur at the right moments. Because these objectives have different physical units, numerical ranges, and dynamical characteristics, and are coupled through the robot dynamics, motion imitation is inherently a multi-objective control problem.

Traditional methods transform several tracking errors separately and combine them into a scalar reward through manually chosen weights and scales. These coefficients determine both the relative importance and the numerical response of the tracking objectives. Because typical errors and dynamical difficulty change with the motion, a coefficient set that serves one skill need not serve another; in practice, reward design is often re-tuned whenever the reference motion changes.

Adversarial imitation offers a way to reduce manual reward design. Instead of declaring in advance how many meters of root-position error are worth how many radians of rotation error, a discriminator learns a unified imitation signal directly from data. AMP and related methods learn rewards that

describe motion style and plausibility by discriminating whether short state transitions come from the reference distribution [5]. For motion tracking that must strictly follow a synchronized reference trajectory, ADD further feeds the reference-policy state differential to the discriminator and asks whether it is close to the ideal zero differential [6]. The aggregation among tracking objectives can then be learned by the discriminator instead of specifying component-wise weights and scales by hand.

Differential adversarial rewards, however, still face an important issue. Their ideal sample is always a single point, the zero differential. Without a suitable constraint, the discriminator easily becomes too sharp: it assigns a high score to the origin and compresses the many nonzero residuals produced by the policy into a nearby low-score region. Even if classification succeeds, the discriminator may not provide an effective continuous learning signal for the policy. Existing methods constrain the discriminator through regularization, but one regularizer simultaneously affects two distinct properties. On the one hand, it controls the discriminator’s local sensitivity, that is, how fast the score changes with the input residual. On the other hand, it also affects the numerical range in which the final scores live, and this range in turn determines whether the policy reward operates in an effective interval. As the residual distribution changes across motions, the same regularization setting can correspond to different sensitivities and different score scales, producing motion-dependent effective regularization requirements.

The core idea of DARE is to treat these two issues separately rather than control them with one motion-dependent coefficient. First, DARE applies spectral normalization to every linear layer of the differential discriminator, directly bounding the maximum amplification of residual variations. Second, after residual normalization is fixed, DARE measures the score separation between the unique zero-error anchor and representative policy residuals and rescales the discriminator once in closed form. The geometry constraint and reward operating scale are thus specified independently.

Constraining the critic’s sensitivity alone does not resolve the score scale. A Lipschitz bound controls how rapidly the score may change, but not the numerical range actually used by the trained critic. DARE therefore measures the anchor-to-typical-residual score gap once after normalization stabilizes and uses a closed-form global rescaling tied to the response range of the reward transform. All motions use exactly the same calibration rule; the only quantity that changes across motions is the residual statistic measured automatically from the training data. The critic’s geometry and score scale thus

no longer depend on the same hand-tuned parameter.

The main contributions are threefold. First, we identify that the regularizer in existing differential adversarial rewards simultaneously affects the critic’s local geometry and its score scale, and that this coupling can make different motions require different effective regularization settings. Second, we propose DARE, which constrains discriminator sensitivity through layer-wise spectral normalization and sets the score scale automatically through one-shot anchor calibration, removing the need for a motion-specific gradient-penalty coefficient. Third, we evaluate one shared DARE configuration on six skills with different durations, contact patterns, and dynamical characteristics.

II. RELATED WORK

Physics-based motion tracking and learned imitation objectives. DeepMimic combines pose, velocity, root, and end-effector tracking terms for physics-based skills [1]; subsequent work improves scalability and motion coverage through architectures, representations, and data [2], [3]. Adversarial imitation provides a learned alternative to a manually written reward: GAIL matches expert and policy occupancies [4], while AMP learns a motion prior by discriminating short state transitions [5]. ADD addresses reference-conditioned tracking by presenting the reference–policy differential to an adversarial discriminator and learning a nonlinear aggregation of tracking objectives without component-wise reward weights [6].

Multi-objective learning and objective aggregation. Competing objectives are commonly combined by weighted scalarization, Pareto optimization, or adaptive loss weighting [10], [11]. These approaches differ in whether preferences are specified explicitly or adapted from objective statistics. In differential adversarial tracking, the discriminator instead learns this aggregation from the joint residual distribution.

Regularization of adversarial critics. Adversarial critics are regularized with gradient penalties or spectral normalization to control sensitivity and improve optimization [7], [8]. Gradient penalties constrain sampled input derivatives, whereas spectral normalization bounds the gain of individual network maps by construction. For non-Wasserstein adversarial losses, the numerical score range also determines the operating regime of the loss and its reward transform [9]. Geometry and score calibration are therefore related but distinct considerations when a critic supplies the learning signal.

III. BACKGROUND

A. Reference-Conditioned Motion Tracking

In reference-conditioned humanoid motion imitation, the controller must keep the executed motion continuously aligned with a time-synchronized reference trajectory. Let s_t and $\hat{s}_t$ denote the current state and the reference state at time t . Tracking quality is jointly measured by several physical quantities, including root position and orientation, articulated body pose, and motion velocities. Let $\phi_i(s_t)$ extract the i -th tracking quantity from the current state and

let $\phi_i(\hat{s}_t)$ denote its reference counterpart. These quantities may have different dimensions and physical units—positions are measured in meters, while rotations and velocities have their own representations.

B. Handcrafted Objective Scalarization

Traditional methods first compress each tracking error into a scalar. For tracked quantity i , the scalar tracking error at time t is the Euclidean distance $e_{i,t} = \|\phi_i(\hat{s}_t) - \phi_i(s_t)\|_2$.

A representative handcrafted scalarization is

$$r_t^{\text{hand}} = \sum_i w_i \exp(-\alpha_i e_{i,t}^2), \quad (1)$$

where w_i and α_i set the relative importance and numerical scale of each tracking quantity. DeepMimic is a representative instance of this class: its tracking terms use weights and error scales specified before training and held fixed throughout [1]. This is a fixed scalarization of the tracking objectives: the trade-off among errors is fixed through constant coefficients chosen before training starts. Because typical error magnitudes and dynamical characteristics differ across motions, one coefficient set need not serve all motions. Learned scalarization instead leaves the trade-off unspecified in advance and learns the aggregation among objectives from the error distribution that actually occurs during training.

C. Differential Adversarial Objectives

Differential adversarial learning provides such a learned aggregation [6]. Consider n non-negative objective losses $l^1, \dots, l^n$ collected into a differential vector $\Delta = (l^1, \dots, l^n)$, each entry recording the gap between the current solution and the ideal on one objective; when all objectives are simultaneously ideal, $\Delta = 0$. A discriminator $D(\Delta)$ is trained to separate the ideal zero differential from the nonzero differentials produced by the current policy. Since the discriminator receives the differentials of all objectives at once, it can learn a nonlinear objective aggregation function without fixed per-objective weights. Its local sensitivity to objective i is described by $\partial D / \partial l^i$: the extent to which the discriminator output responds when objective i changes slightly near the current differential. Unlike fixed scalarization weights, this local sensitivity can vary with the position in the residual space and during training. In synchronized motion tracking, the differential is no longer restricted to one scalar loss per objective but can be formed directly from the feature residual between the current and reference states. Perfect tracking again corresponds to the unique zero differential, while nonzero differentials retain the error direction and the relations among features. The policy reward is generated from the discriminator’s score on the current differential, for example $r_t = -\log(1 - D(\Delta_t))$: the closer the current residual is judged to the ideal state, the higher the reward.

IV. DARE: DIFFERENTIAL ADVERSARIAL REWARD EQUALIZATION

A. Learned Differential Reward

Motion tracking involves multiple objectives with different physical meanings. Rather than compressing them into sepa-

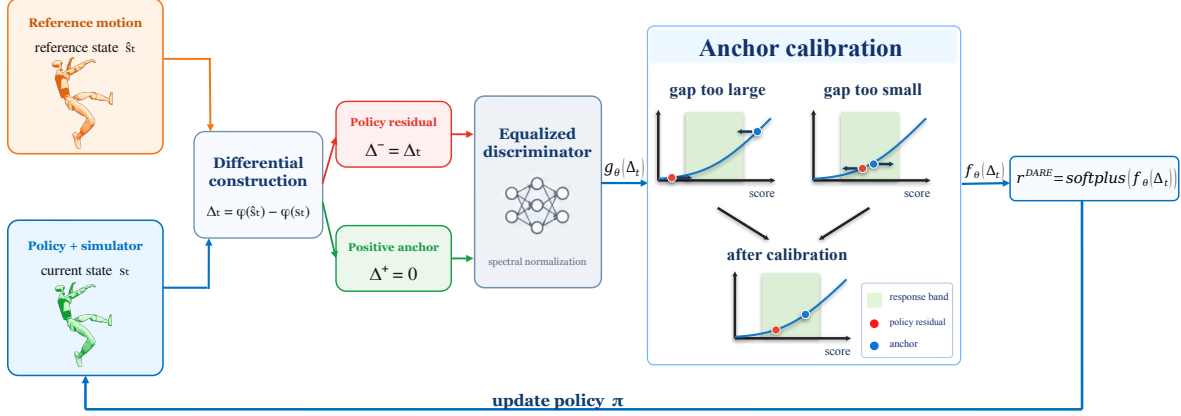


Fig. 1. DARE training loop. The synchronized reference and simulated state form the policy residual. The residual and zero anchor train the equalized discriminator; one-shot anchor calibration rescales its logit, and the resulting softplus reward drives the PPO update.

rately weighted scalar terms, DARE forms one full reference–policy residual. Let $\phi(s_t) \in \mathbb{R}^d$ collect the tracked quantities and let $\nu \in \mathbb{R}^d$ contain their per-coordinate mean-absolute scales, estimated during early training and fixed after the first 10^8 samples. The normalized differential is $\Delta_t = \delta_t \oslash \nu$, where $\delta_t = \phi(\hat{s}_t) - \phi(s_t)$, and perfect tracking is represented by $\Delta^+ = 0$. Here $\oslash$ denotes element-wise division. Because normalization only rescales coordinates and introduces no offset, perfect tracking remains the unique zero differential. Rotational quantities use a tangent–normal representation so that all tracked residuals are represented in consistent Euclidean coordinates.

The full differential retains coordinate-wise error magnitudes, directions, and correlations. Two states can therefore have the same aggregate distance while differing in which tracked quantities deviate from the reference. Let q_π denote the distribution of differentials induced by policy π . During training, recently generated policy differentials are stored in a replay buffer and reused as negative samples for critic updates.

DARE maps the full differential to a scalar discriminator logit, $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$. The corresponding discriminator output is $D_\theta(\Delta) = \sigma(f_\theta(\Delta)) \in (0, 1)$. A larger $D_\theta(\Delta)$ indicates that the differential is classified as more likely to belong to the positive, perfect-tracking class. Perfect tracking defines the positive differential $\Delta^+ = 0$, while differentials induced by policy π form the negative class, with $\Delta^- \sim q_\pi$. The discriminator is trained to distinguish these two classes:

$$\max_{\theta} \log D_\theta(\Delta^+) + \mathbb{E}_{\Delta^- \sim q_\pi} [\log(1 - D_\theta(\Delta^-))] \quad (2)$$

The first term encourages a high discriminator output for the positive differential Δ^+ , while the second encourages low outputs for policy-induced negative differentials Δ^- . As the policy changes, the residuals used to train the discriminator

also change. Because the discriminator receives all residual coordinates together and produces a single logit, it learns how the different tracking errors contribute to the final score.

To train the discriminator by loss minimization, we use the balanced negative log-likelihood corresponding to the binary classification objective in Eq. (2). Using $-\log \sigma(z) = \text{softplus}(-z)$ and $-\log(1 - \sigma(z)) = \text{softplus}(z)$, the discriminator loss becomes

$$\mathcal{L}_{\text{DARE}}(\theta) = \frac{1}{2} \text{softplus}[-f_\theta(\Delta^+)] + \frac{1}{2} \mathbb{E}_{\Delta^- \sim q_\pi} \text{softplus}[f_\theta(\Delta^-)] \quad (3)$$

This objective provides only two types of supervision: the positive differential $\Delta^+ = 0$ representing perfect tracking and the policy-induced negative differentials $\Delta^- \sim q_\pi$. The discriminator therefore learns from data how the different tracking components should jointly determine its logit.

The policy then constructs its reward directly from the discriminator logit. DARE defines

$$r_t^{\text{DARE}} = -\log[1 - D_\theta(\Delta_t)] = \text{softplus}[f_\theta(\Delta_t)] \quad (4)$$

Since the softplus function is strictly increasing, a larger discriminator logit produces a larger policy reward. The discriminator is trained to assign a high output to the positive differential Δ^+ and lower outputs to policy-induced negative differentials Δ^- , so the resulting reward encourages the policy toward states that the discriminator judges more consistent with perfect tracking. The policy can thus improve root motion, body pose, and velocities simultaneously through one unified scalar reward.

B. Equalized Discriminator: Sensitivity Constrained by Construction

The differential discriminator must not only distinguish perfect tracking from policy-induced residuals, but also

provide the policy with a continuous reward signal that reflects tracking quality. Therefore, the sensitivity of the discriminator to the tracking differential must be controlled. Since perfect tracking always corresponds to the unique zero differential $\Delta^+ = 0$, excessive sensitivity can cause the discriminator to form an overly steep response near the origin, so that the logit drops rapidly once the residual deviates only slightly from zero. Although such a discriminator can still separate positive and negative samples, it may map nonzero residuals of clearly different quality to a similarly low-score region, thereby weakening the ability of the reward to distinguish different tracking qualities.

A common approach is to introduce a gradient penalty during discriminator training to suppress excessive input sensitivity [6]. However, this introduces an additional regularization strength: if the constraint is too weak, the discriminator may remain overly sensitive; if it is too strong, the discriminator may become overly smooth and lose the ability to distinguish different residuals. DARE therefore constrains the maximum sensitivity of the discriminator directly through the network architecture, without introducing this additional balancing parameter.

The normalized full tracking differential $\Delta \in \mathbb{R}^d$ is fed directly into the discriminator network. Let the network contain L linear layers, and let the weight of the ℓ -th layer be W_ℓ . DARE applies spectral normalization to each linear layer, i.e., $\bar{W}_\ell = W_\ell / \sigma_{\max}(W_\ell)$, where $\sigma_{\max}(W_\ell)$ denotes the maximum amplification that this linear map can apply to an input variation. After normalization, $\sigma_{\max}(\bar{W}_\ell) = 1$. The network uses ReLU as the activation function ρ , and ReLU is also a 1-Lipschitz mapping. Therefore, each linear layer together with its activation does not amplify an input variation by more than a factor of one.

Let the discriminator logit before scale calibration be $g_\theta(\Delta)$. Since the entire discriminator is a continuous composition of these non-expansive mappings, its Lipschitz constant satisfies

$$\text{Lip}(g_\theta) \leq \prod_{\ell=1}^L \sigma_{\max}(\bar{W}_\ell) = 1. \quad (5)$$

Therefore, for any two tracking differentials Δ and Δ' , we have

$$|g_\theta(\Delta) - g_\theta(\Delta')| \leq \|\Delta - \Delta'\|_2. \quad (6)$$

This means that the discriminator cannot arbitrarily amplify a very small change in the tracking residual into a very large change in its logit; its maximum sensitivity is directly constrained by the network architecture.

Since perfect tracking corresponds to $\Delta^+ = 0$, setting $\Delta' = 0$ in Eq. (6) further gives

$$|g_\theta(\Delta) - g_\theta(0)| \leq \|\Delta\|_2. \quad (7)$$

Therefore, when the tracking differential is close to the perfect-tracking anchor, its raw discriminator logit cannot change arbitrarily rapidly relative to the logit at zero residual. DARE thus directly constrains the maximum response rate of the discriminator to the tracking residual through layer-wise spectral normalization of the discriminator network.

C. Anchor Calibration: Closed-Form Output Scale

Spectral normalization constrains the geometric sensitivity of the raw discriminator, but it does not determine the global logit scale that is ultimately used. This scale directly affects how the reward responds to changes in tracking quality. DARE uses $r^{\text{DARE}} = \text{softplus}(f_\theta)$ to convert the calibrated discriminator logit f_θ into the policy reward, so the slope of the reward with respect to the logit is

$$\frac{\partial r^{\text{DARE}}}{\partial f_\theta} = \sigma(f_\theta). \quad (8)$$

When f_θ is a small negative number, $\sigma(f_\theta)$ is close to 0, and a change in the logit can only produce a very small difference in reward; near $f_\theta = 0$, the reward is more sensitive to changes in the logit; as f_θ increases, the slope gradually approaches 1. Therefore, in addition to limiting the sensitivity of the discriminator, its logits must be kept within a numerical range in which softplus retains sufficient response capability.

DARE takes the logit interval over which the reward slope grows from 0.2 to 0.8 as the main response region. Since the reward slope is exactly $\sigma(f_\theta)$, the width of this interval is

$$w = \sigma^{-1}(0.8) - \sigma^{-1}(0.2) = \ln 16 \approx 2.77. \quad (9)$$

Thus, $w = \ln 16$ gives a target logit separation determined by the reward transform itself.

To match the actual output scale of the discriminator to this target range, DARE uses the fixed zero differential $\Delta = 0$ provided by perfect tracking as the calibration anchor. Let the discriminator output before scale calibration be $g_\theta(\Delta)$, so the raw logit corresponding to perfect tracking is $g_\theta(0)$. Meanwhile, the nonzero differentials in the current rollout and the replay buffer represent the tracking residuals actually produced by the current policy. DARE describes the position of the current policy residuals in logit space by their mean raw logit, and measures its separation from the zero anchor at calibration time t_f :

$$M_f = g_\theta(0) - \frac{1}{2} \left[\mathbb{E}_{\Delta \sim q_{\text{cur}}} g_\theta(\Delta) + \mathbb{E}_{\Delta \sim q_{\text{rep}}} g_\theta(\Delta) \right] \Bigg|_{t=t_f}. \quad (10)$$

Here, q_{cur} and q_{rep} denote the distributions of residual differentials in the current rollout and the replay buffer, respectively. Thus, M_f measures the actual separation between perfect tracking and policy residuals in raw logit space at the calibration time.

Given the target separation w and the measured separation M_f , DARE uniformly rescales the entire discriminator logit through a global scale factor:

$$f_\theta(\Delta) = \kappa g_\theta(\Delta), \quad \kappa = \frac{w}{M_f}. \quad (11)$$

When $M_f > w$, we have $\kappa < 1$, and an overly wide logit separation is compressed; when $M_f < w$, we have $\kappa > 1$, and an overly narrow separation is enlarged. Since all logits are multiplied by the same positive scale factor, this operation preserves the score ordering learned by the discriminator and only adjusts its overall output scale.

Algorithm 1 DARE Training

Require: reference motion $\mathcal{M}$, policy π_φ , value V_ω , raw critic g_θ , replay buffer $\mathcal{B}$, normalizer $\mathcal{N}$; $\kappa \leftarrow 1$

- 1: **for** iteration $t = 1, 2, \dots$ **do**
- 2: Collect a synchronized rollout with π_φ from random reference phases; construct $\Delta_t = \delta_t \odot \nu$; add to $\mathcal{B}$
- 3: **if** $\mathcal{N}$ was fixed this iteration **then**
- 4: $\bar{g}_{\text{cur}} \leftarrow \mathbb{E}_{\Delta \sim q_{\text{cur}}} g_\theta(\Delta)$
- 5: $\bar{g}_{\text{rep}} \leftarrow \mathbb{E}_{\Delta \sim q_{\text{rep}}} g_\theta(\Delta)$
- 6: $M_f \leftarrow g_\theta(0) - \frac{1}{2}(\bar{g}_{\text{cur}} + \bar{g}_{\text{rep}})$; $\kappa \leftarrow \ln 16/M_f$
- 7: **end if**
- 8: Compute r_t^{DARE} by (4) with $f_\theta = \kappa g_\theta$
- 9: **for** critic minibatch **do**
- 10: Sample equal numbers of current and replay differentials; update g_θ by one step on (3) with $f_\theta = \kappa g_\theta$
- 11: **end for**
- 12: Update π_φ and V_ω with PPO on r_t^{DARE}
- 13: **end for**

Calibration is performed once after input normalization is fixed. At this point, the numerical scale of the differential coordinates is already stable, so the measured M_f has a fixed scale meaning. We keep $\kappa = 1$ before calibration; after obtaining κ , we fix it and use the calibrated logit $f_\theta = \kappa g_\theta$ consistently in subsequent discriminator optimization and policy reward computation.

Ultimately, the target width $w = \ln 16$ is determined directly by the response characteristics of the softplus reward, while M_f is measured from the current training distribution. Therefore, the calibration scale in (11) is obtained directly in closed form, without introducing an additional tunable reward-scale parameter.

D. Training

Training alternates rollout collection, differential construction, critic updates, and PPO updates (Algorithm 1). Critic minibatches draw an equal mixture of current and replay differentials, with the ideal point presented once per minibatch as the positive class; at t_f , before the rewards of that iteration are computed, M_f is measured with the critic in evaluation mode and κ is set by (11). One critic, one reward, one policy, and one calibration rule make up the whole loop. Figure 1 summarizes this information flow from the reference and policy states to the calibrated reward and PPO update.

V. EXPERIMENTS

A. Experimental Setup

All methods are evaluated using the same 28-DoF, 15-body humanoid model adopted in prior physics-based motion-imitation studies, simulated in Isaac Lab at 120Hz with actions applied at 30Hz through PD joint-position control [14], [15]. All four methods share the same environment, training budget, motion clips, random-phase initialization, and evaluation protocol. Each policy is trained for 2000 iterations of $8192 \text{ environments} \times 32 \text{ steps}$ (5.24×10^8 samples) with five independent seeds, and is evaluated on 4096 episodes per seed from randomly sampled reference phases. Actor and value networks are two-layer MLPs (1024,

TABLE I

DARE CONFIGURATION SHARED ACROSS ALL SIX SKILLS. THE CALIBRATION RULE IS SHARED; THE RESULTING κ IS DATA-DEPENDENT.

Component	Setting
Differential input	single frame, $d = 172$
Discriminator	MLP, $172 \rightarrow 1024 \rightarrow 512 \rightarrow 1$
Activation ρ	ReLU (1-Lipschitz)
Spectral norm.	all linear maps, one power iteration
Normalizer $\mathcal{N}$	per-coordinate mean-abs scale, fixed after 10^8 samples (iter. 382)
Logit calibration	$\kappa = \ln 16/M_f$, once at iter. 382
Critic optimizer	SGD, lr 2.5×10^{-4} , momentum 0.9
Critic minibatch	16384 current + 16384 replay
Replay buffer	2×10^5 differentials

512), and the PPO settings (clip 0.2, GAE 0.95, $\gamma = 0.99$, SGD with momentum 0.9) are fixed across methods. DARE additionally uses the normalizer and one-shot logit calibration specified in Table I.

The benchmark contains six skills with substantially different motion durations, contact patterns, and dynamical characteristics: Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick. Training termination follows each method’s environment configuration (pose termination is enabled for DeepMimic, ADD, and DARE and disabled for AMP). The evaluation harness accumulates each metric over the executed reference-aligned steps and uses the same aggregation rule for every method; terminated trajectories are not padded with synthetic states. Final policies reach essentially the full prescribed evaluation horizon, so early termination has negligible effect on the reported averages.

We use six complementary measures. The four geometric measures quantify frame-wise imitation fidelity, whereas foot progress and lateral drift capture trajectory-level behavior that is not summarized by instantaneous pose errors. Four describe geometric tracking at two spatial levels: root position and root rotation measure global translation and orientation, while body position and body rotation measure articulated-body tracking. For an evaluation trajectory with B bodies, where r, p_b and q, q_b denote simulated root, body positions and orientations and hats denote reference values, these quantities are

$$\begin{aligned}
 E_{\text{root-pos}} &= \|\hat{r} - r\|_2, & E_{\text{root-rot}} &= \angle(q, \hat{q}), \\
 E_{\text{body-pos}} &= \frac{1}{B} \sum_{b=1}^B \|(\hat{p}_b - \hat{r}) - (p_b - r)\|_2, \\
 E_{\text{body-rot}} &= \frac{1}{B} \sum_{b=1}^B \angle(q_b, \hat{q}_b).
 \end{aligned} \tag{12}$$

where $\angle(\cdot, \cdot)$ is the shortest angle between normalized quaternions. Geometric quantities are evaluated at each reference-aligned timestep and then averaged over executed timesteps and episodes; the two behavior measures are computed per trajectory and then averaged over episodes. The remaining two measures describe motion-level behavior. Let Δr and $\Delta \hat{r}$ be the simulated and reference root displacements over the evaluated trajectory, and let $\Delta p_j, \Delta \hat{p}_j$ be the corresponding

TABLE II

MACRO-AVERAGE TRACKING AND MOTION-LEVEL BEHAVIOR UNDER THE MATCHED TRAINING AND EVALUATION PROTOCOL. FOR EACH SEED, EACH METRIC IS MACRO-AVERAGED EQUALLY OVER THE SIX SKILLS; ENTRIES REPORT THE MEAN $\pm$ ONE STANDARD DEVIATION ACROSS FIVE SEEDS. LOWER IS BETTER EXCEPT FOR FOOT PROGRESS, WHICH IS IDEAL AT ONE. DARE USES THE SAME CONFIGURATION ON ALL SKILLS.

(a) Geometric tracking				
Metric $\downarrow$	AMP	DeepMimic	ADD	DARE
Root Pos. [m]	0.9116 $\pm$ 0.084	0.0703 $\pm$ 0.012	0.0725 $\pm$ 0.010	0.0528$\pm$0.008
Body Pos. [m]	0.2169 $\pm$ 0.031	0.0557 $\pm$ 0.009	0.0363 $\pm$ 0.007	0.0313$\pm$0.006
Root Rot. [rad]	0.7311 $\pm$ 0.076	0.1459 $\pm$ 0.021	0.0673 $\pm$ 0.011	0.0529$\pm$0.009
Body Rot. [rad]	0.8098 $\pm$ 0.069	0.2145 $\pm$ 0.028	0.1358 $\pm$ 0.017	0.1246$\pm$0.013
(b) Motion-level behavior				
Metric	AMP	DeepMimic	ADD	DARE
Foot progress $\rightarrow$ 1	1.1335 $\pm$ 0.046	1.0060 $\pm$ 0.018	1.0109 $\pm$ 0.021	0.9989$\pm$0.012
Lateral drift $\downarrow$	0.1127 $\pm$ 0.019	0.0277 $\pm$ 0.007	0.0435 $\pm$ 0.009	0.0171$\pm$0.005

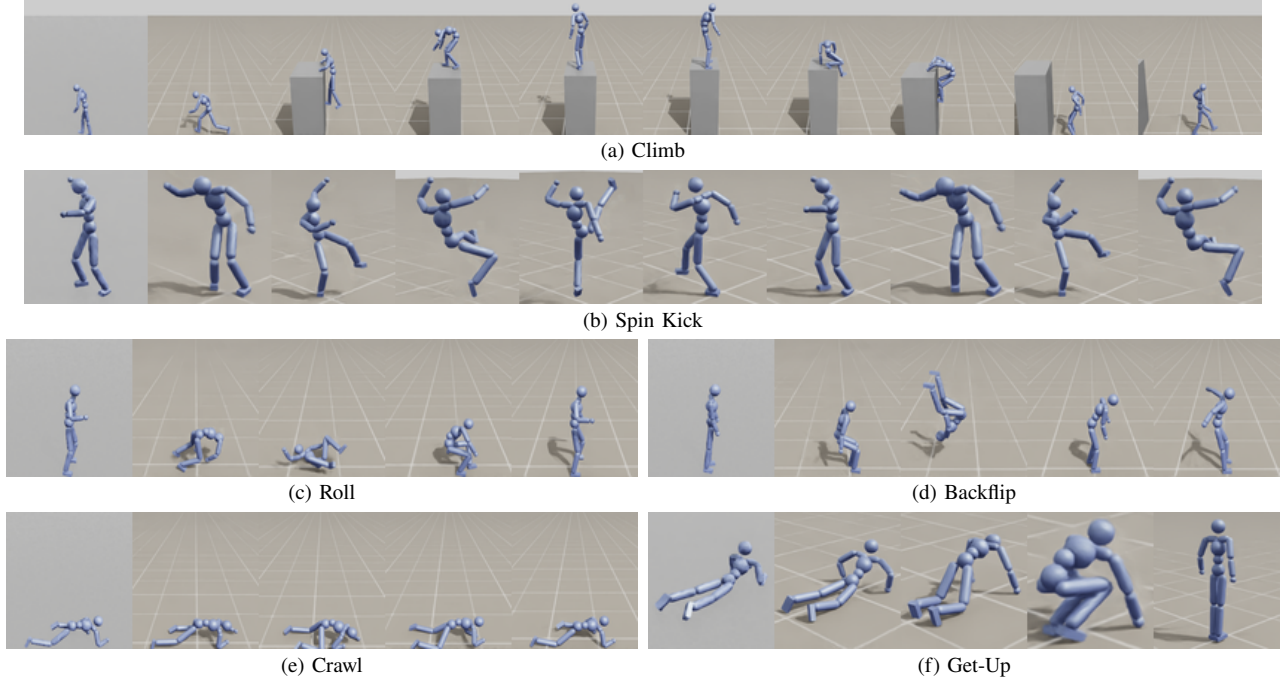


Fig. 2. DARE policies executing the six skills in Isaac Lab using the shared configuration in Table I. Frames are uniformly sampled from each rollout and rendered from a fixed front-view camera. Climb and Spin Kick are shown with ten frames, and the remaining skills with five.

displacements of foot j . The reference-projected progress and transverse drift are

$$\begin{aligned}
 \rho_{\text{foot}} &= \min_j \frac{\Delta p_j^\top \Delta \hat{p}_j}{\|\Delta \hat{p}_j\|_2^2}, \\
 \rho_r &= \frac{\Delta r^\top \Delta \hat{r}}{\|\Delta \hat{r}\|_2^2}, \\
 E_{\text{lat}} &= \frac{\|\Delta r - \rho_r \Delta \hat{r}\|_2}{\|\Delta \hat{r}\|_2}.
 \end{aligned} \tag{13}$$

Foot progress is ideal at one, while lateral drift is minimized at zero. All six quantities are averaged over the evaluation trajectories; lower is better except for foot progress, whose target value is one.

B. Cross-Motion Tracking

Table II asks whether one configuration can serve skills whose contacts, horizons, and velocity profiles differ, and

Fig. 2 shows the resulting controllers executing the six skills in Isaac Lab. With one configuration shared across all six skills, DARE achieves the best macro-average on all six measures. It yields the lowest root- and body-level position and rotation errors, the foot-progress ratio closest to one, and the lowest lateral drift. Relative to the strongest baseline for each measure, root-position error, root-rotation error, and lateral drift are reduced by approximately 25%, 21%, and 38%.

C. Learning Dynamics

Figure 3 compares the five-seed mean body-position and root-rotation errors against environment samples for Roll and Climb, which provide contrasting motion durations and contact structures. Across both, the DARE curves converge to lower errors under the same sample budget, complementing the final-checkpoint comparison in Table II.

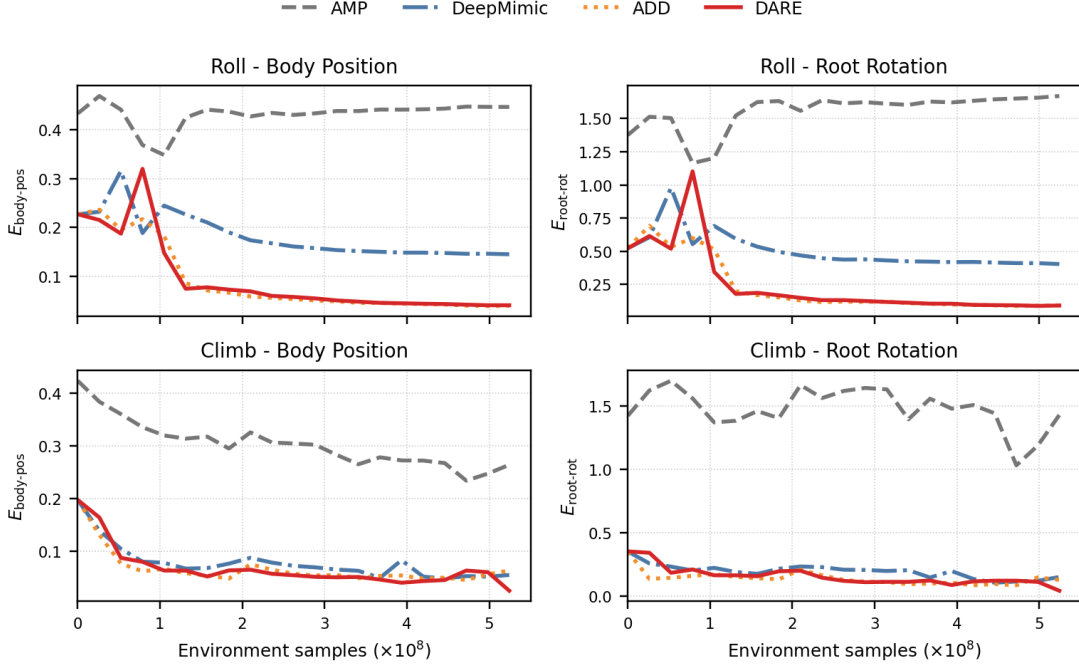


Fig. 3. Learning curves for Roll and Climb under the matched training budget. Curves show the mean over five independent seeds. The four panels report body-position and root-rotation errors versus environment samples, representing articulated translational fidelity and global orientation tracking, respectively. Lower is better.

D. Component Analysis

We isolate the two components that define DARE’s critic: layer-wise spectral normalization and one-shot score calibration. This gives a 2×2 comparison while keeping the differential input, binary objective, replay procedure, PPO budget, and evaluation protocol fixed. The study covers eight motions—Roll, Climb, Backflip, Crawl, Get-Up, Spin Kick, Push Box, and Sit on Sofa—with three independent seeds for every configuration. Entries are macro-averaged over the eight motions for each seed and then summarized as a mean $\pm$ standard deviation; cells whose measurements are still pending are marked in red.

TABLE III

COMPONENT ANALYSIS OVER EIGHT MOTIONS. SN AND CAL. DENOTE LAYER-WISE SPECTRAL NORMALIZATION AND ONE-SHOT CALIBRATION. LOWER IS BETTER; RED ENTRIES ARE PENDING MEASUREMENTS.

Method	SN	Cal.	R.Pos. ↓	B.Pos. ↓	R.Rot. ↓	B.Rot. ↓	DoF Vel. ↓
DARE	✓	✓	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
w/o SN	×	✓	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
w/o Cal.	✓	×	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Base	×	×	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]

VI. CONCLUSION

Manual tracking rewards turn the aggregation of heterogeneous imitation objectives into per-motion engineering. Learned differential rewards remove the component-wise weights, but regularization can still couple the critic’s geometric sensitivity to the score range seen by the policy.

DARE treats these as separate design quantities: layer-wise spectral normalization constrains the critic’s sensitivity, and one closed-form calibration sets the score scale without a motion-specific coefficient. The score change relative to the unique zero differential is bounded by the normalized residual, and the critic is then rescaled once, in closed form, to set its operating scale relative to the response width of the reward transform. Across the six skills, DARE attains the lowest macro-average for all four geometric errors, the foot-progress ratio closest to one, and the smallest lateral drift in Table II. The formulation also provides a direct path toward multi-clip training and other morphologies, where additional objectives can be incorporated directly into the differential.

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